German University in Cairo

Faculty of Information Engineering and Technology (IET)
Dr. Sarah Azzam
Eng. Alaa Taha
Eng. Amr El Kamash
Eng. Omar Abd El-Rahim

CSIS 402 Computer Organization & System Programming

Project Description – Milestone 1

An 8086 Implementation of a 128bit Advanced Encryption Standard (AES)

About AES

The Advanced Encryption Standard (AES) is a symmetric-key (private-key) block cipher standardized by NIST in 2001, based on the Rijndael algorithm used to protect classified information. It encrypts data in 128-bit blocks, represented as a 4x4 matrix of bytes called the state. With a 128-bit key, AES performs 11 rounds: an initial AddRoundKey, then 9 rounds of SubBytes, ShiftRows, MixColumns and AddRoundKey, and a final round without MixColumns. Each round key is derived from the original key through the key expansion process.

The complete specification (FIPS 197):
http://csrc.nist.gov/publications/fips/fips197/fips-197.pdf

Animation of the algorithm (recommended viewing before starting):
https://formaestudio.com/rijndaelinspector/archivos/Rijndael_Animation_v4_eng-html5.html

MixColumns is a bit tough and needs extra work; its clear description is available in this document. Try to start with others first to get a better feeling:
https://drive.google.com/file/d/1Rcg_XGZlg6KBsE8neJ7r0nVq66LX3jDA/view?usp=drive_link

Team Submission

- Team size: 2 students (can be from different tutorials/majors)
- **Deadline:** **Saturday 7/11**
- Submission link: https://forms.gle/fRdKpDMMjtMgFCqo6

Milestone 1 Requirements

Deadline: **Saturday 28/11**.

Write assembly macros for the four AES round operations, combined into one full round of encryption as follows:

1- Use Macros to implement SubBytes() and ShiftRows() modules, all work on 128 bits.
2- Use Macros to implement one cycle of MixColumns() and AddRoundKey() modules, all work on 128 bits.
3- For the AddRoundkey module consider the used key of F F F F F F F F F F F F F F F F.

Remarks

- Using EMU8086 or any other 8086 emulator for this project is acceptable.
- The evaluation method and Milestone 2 will be announced later.